

#Practise Question

1. Employee Salary Protection

Create a class Employee:

Private attribute: salary

Methods:

set_salary(amount) → validate salary > 0

get_salary() → return salary.

2. Bank Account System

Create a BankAccount class:

Private attributes: balance, account_number

Methods:

deposit(amount)

withdraw(amount) (check insufficient balance)

get_balance()

3. Student Grade System

Create a Student class:

Private attribute: grades (list)

Methods:

add_grade(mark)

get_average()

4. E-commerce Product Pricing

Create a Product class:

Private attribute: price

Methods:

set_price(price) → price cannot be negative

apply_discount(percent)

get_price()

5. User Login System

Create a User class:

Private attributes: username, password

Methods:

set_password(new_password) (min length validation)

check_password(input_password)

6. Car Speed Control System

Create a Car class:

Private attribute: speed

Methods:

accelerate(value)

brake(value)

get_speed()

7. ATM Machine Simulation

Create an ATM class:

Private attributes: pin, balance

Methods:

verify_pin(input_pin)

withdraw(amount, pin)

deposit(amount)

check_balance(pin)

8. Hospital Patient Record System

Create a Patient class:

Private attributes: medical_history, diagnosis

Methods:

add_record(record)

view_records(authorized_user)

9. Inventory Management System

Create an InventoryItem class:

Private attributes: stock_quantity

Methods:

add_stock(qty)

remove_stock(qty)

get_stock()

10. Online Course Platform

Create a Course class:

Private attributes: enrolled_students

Methods:

enroll(student)

unenroll(student)

get_total_enrollment()

11. Payment Gateway System

Create a Payment class:

Private attributes: card_number, cvv, balance

Methods:

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process_payment(amount)
validate_card()
get_masked_card() (show only last 4 digits
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